

Problem

Find the Laplace transform of $f(t) = 15 \cos(3t)$ using the exponential representation for the cosine function.

Step-by-step solution

Step 1 of 2

Determine the Laplace transform of $f(t)$ using equation

$$L[f(t)] = \int_0^{\infty} f(t) e^{-st} dt$$

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

Substitute $15 \cos(3t)$ for $f(t)$.

$$\begin{aligned} F(s) &= \int_0^{\infty} 15 \cos(3t) e^{-st} dt \\ &= 15 \int_0^{\infty} \cos(3t) e^{-st} dt \end{aligned}$$

Step 2 of 2

Represent $\cos \omega t$ signal in terms of exponential signal.

$$\cos \omega t = \frac{e^{j\omega t} + e^{-j\omega t}}{2}$$

Substitute 3 for ω .

Using exponential equation for cosine function,

$$\cos(3t) = \frac{e^{j3t} + e^{-j3t}}{2}$$

Substitute $\frac{e^{j3t} + e^{-j3t}}{2}$ for $\cos(3t)$.

$$\begin{aligned} F(s) &= 15 \int_0^{\infty} \left(\frac{e^{j3t} + e^{-j3t}}{2} \right) e^{-st} dt \\ &= \frac{15}{2} \left[\int_0^{\infty} e^{j3t} e^{-st} dt + \int_0^{\infty} e^{-j3t} e^{-st} dt \right] \\ &= \frac{15}{2} \left[\int_0^{\infty} e^{-(s-j3)t} dt + \int_0^{\infty} e^{-(s+j3)t} dt \right] \\ &= \frac{15}{2} \left[\left[\frac{e^{-(s-j3)t}}{-(s-j3)} \right]_0^{\infty} + \left[\frac{e^{-(s+j3)t}}{-(s+j3)} \right]_0^{\infty} \right] \end{aligned}$$

$$\begin{aligned} F(s) &= \frac{15}{2} \left[\left[\frac{0}{-(s-j3)} - \frac{1}{-(s-j3)} \right] + \left[\frac{0}{-(s+j3)} - \frac{1}{-(s+j3)} \right] \right] \\ &= \frac{15}{2} \left[\frac{1}{(s-j3)} + \frac{1}{(s+j3)} \right] \\ &= \frac{15}{2} \left[\frac{s+j3+s-j3}{(s-j3)(s+j3)} \right] \\ &= \frac{15}{2} \left[\frac{2s}{s^2+9} \right] \end{aligned}$$

$$F(s) = \frac{15s}{s^2+9}$$

Therefore, the Laplace transform of the signal $15 \cos(3t)$ is $\boxed{\frac{15s}{s^2+9}}$.